

CELLCHECK · DESIGN PRECIS

CellCheck design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, grading method, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

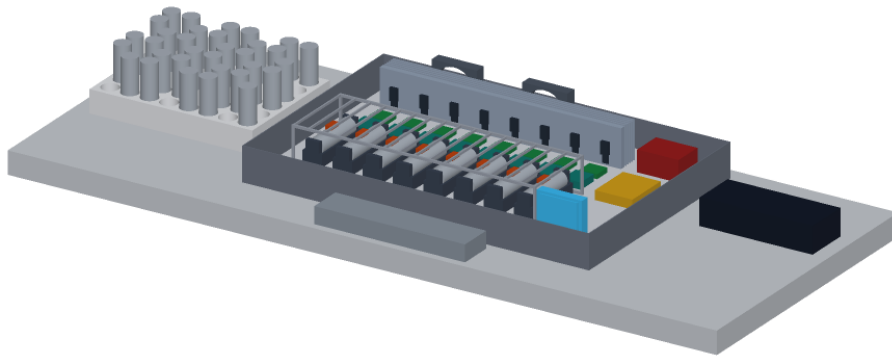
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■ Summary

CellCheck is an eight-channel bench grader for salvaged 18650 and 21700 lithium-ion cells. Each channel charges one cell, measures its DC internal resistance with a two-step current pulse, discharges it at a constant 1 A to measure capacity, and recharges it for a 14-day self-discharge rest. Software then sorts the passing cells into grades and builds matched groups for a pack of a chosen series and parallel count, with a record for every cell. First-order numbers suggest about 6.5 h per cell, about 12 cells per attended 10 h day and about \$159 in parts, just inside the \$160 budget. Every figure below is an estimate for concept review and will be checked at TRL 3.

CellCheck

CONCEPT, NOT FOR FABRICATION



Grey: Bench top, salvaged laptop pack for scale

Figure 1. Concept massing model on a bench: steel tray with eight cell channels (center), rest rack for the self-discharge check (left), 12 V adapter (right) and a salvaged laptop pack (front) for scale.

■ How it works

1. **Intake (by hand).** The operator removes cells from a pack, rejects any with dents, torn wraps, leaks or corrosion, measures open-circuit voltage and rejects cells below 2.0 V. Each cell gets an ID label. Cells below 2.0 V are not revived: deep discharge can grow copper dendrites that cause internal shorts when the cell is charged again.

2. **Charge.** The cell goes into a channel. A 1 A CC-CV charger module brings it to 4.20 V and stops at 0.1 A. A thermistor clipped to the cell stops the channel if the cell passes 45 °C or rises more than 8 K above the bench.
3. **Rest and resistance.** After a 30 min rest the channel runs a DC resistance test in the pattern of IEC 61960: 0.5 A for 10 s, then 2.0 A for 1 s, with resistance taken as the change in voltage over the change in current ([test pattern summarized by Arbin Instruments](#)). IEC 61960 uses 0.2C and 1C; CellCheck uses fixed currents so results are comparable within a batch, not with data sheet values.
4. **Capacity.** A linear constant-current load discharges the cell at 1.0 A to 2.80 V while a current monitor integrates ampere-hours and watt-hours. Cell temperature is logged throughout; a cell that heats abnormally is flagged.
5. **Recharge for rest.** The charger brings the cell back to 4.10 V. The operator moves it to the printed rest rack, which holds 48 cells.
6. **Self-discharge check.** After 14 days the operator puts the cell back into any channel for a 10 s open-circuit reading. A drop of more than 50 mV marks the cell as rejected.
7. **Grade and match.** Software assigns a grade from capacity, resistance, heating and self-discharge (Table 2), then builds groups for the pack the user wants (for example 4S6P): cells of one grade are distributed with a serpentine sort so that each parallel group's total capacity is within ± 1 % of the mean. Serpentine sorting of retired 18650 cells has been studied as a grouping method ([Xi et al., Batteries, 2026](#)), and clustering by measured capacity and resistance produced a markedly better second-life pack than random selection in a recent study ([Olivero-Ortiz et al., PLOS One, 2026](#)).
8. **Record.** Every cell gets a CSV row: ID, source pack, cell model if known, arrival voltage, capacity, resistance, temperature rise, self-discharge, grade, group and dates. Rejected cells are logged too and go to a battery collection point for recycling.

CellCheck: material flow per 100 salvaged cells (all values are estimates)

CONCEPT, NOT FOR FABRICATION

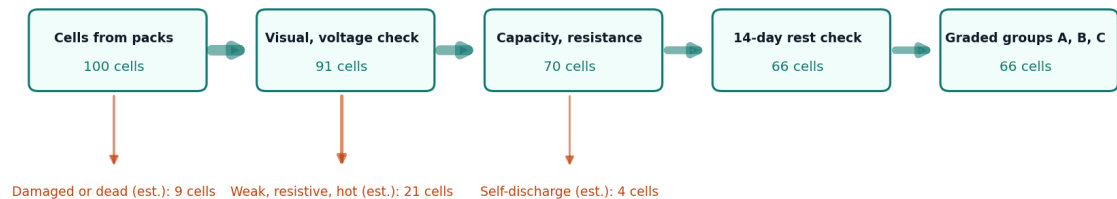


Figure 2. Material flow per 100 cells taken from salvaged packs (all values are estimates). The first loss draws on the 4.3 % of cells with surface damage found by [Salinas et al. \(2019\)](#) plus an assumed 5 % below 2.0 V; the other losses are assumptions to be checked with real cells.

■ Main components

Table 1. Main components, numbered as in the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	Steel tray with liner	460 x 300 x 45 mm steel tray, ceramic fibre sheet on the floor	Contains ejecta from a venting cell; the whole unit sits in it
2	Base plate	Aluminium or FR4 carrier on four standoffs	Carries all modules; standoffs keep it off the tray
3	Cell holders	Eight holders for 18650 and 21700 with spring contacts, separate force and sense contacts at each end (four-wire)	Kelvin sensing keeps contact resistance out of the reading
4	Temperature sensing	10 k Ω NTC thermistor clipped to each cell	Read by the controller's ADC
5	Cell guard	Perforated steel cover over the cells	Stops loose ejecta; cells stay visible
6	Chargers	1 A CC-CV buck charger module per channel, 4.20 V, fed from 12 V (TP5100 class)	Module sets its own voltage limit, independent of firmware
7	Sense and switch boards	INA226-class current and voltage monitor with a 25 m Ω shunt, plus a charger enable switch, per channel	16-bit readings over I2C at addresses 0x40 to 0x47
8	Load MOSFETs	Logic-level MOSFET in TO-220 with an op-amp loop per channel, constant current 0.5 to 2.0 A	Linear load; energy ends up as heat
9	Heatsink	Finned aluminium bar, about 320 mm long	All eight MOSFETs on one bar
10	Fans	Two 60 mm 12 V fans	Temperature-controlled
11	Controller	ESP32-S3 class board	Runs the test sequence, logs to microSD, serves a local web page
12	Display and buttons	2.8 in display with three buttons	Channel status without a laptop
13	Power inlet	DC jack, 6.3 A main fuse, switch	Only 12 V DC enters the unit
14	Power supply	Certified 12 V 5 A mains adapter	No mains wiring inside CellCheck
15	Rest rack	3D-printed rack for 48 cells, numbered slots	Holds cells for the 14-day check

CellCheck: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Steel tray with ceramic fibre liner
- 2 Base plate on standoffs
- 3 Cell holders, four-wire contacts
- 4 NTC thermistor clips
- 5 Perforated steel cell guard
- 6 Charger modules, 1 A CC-CV
- 7 Channel sense and switch boards
- 8 Load MOSFETs
- 9 Heatsink bar
- 10 Fans, 60 mm
- 11 Controller, ESP32-S3 class
- 12 Display and buttons
- 13 Power inlet, fuse and switch
- 14 12 V 5 A certified adapter
- 15 Self-discharge rest rack, 48 cells

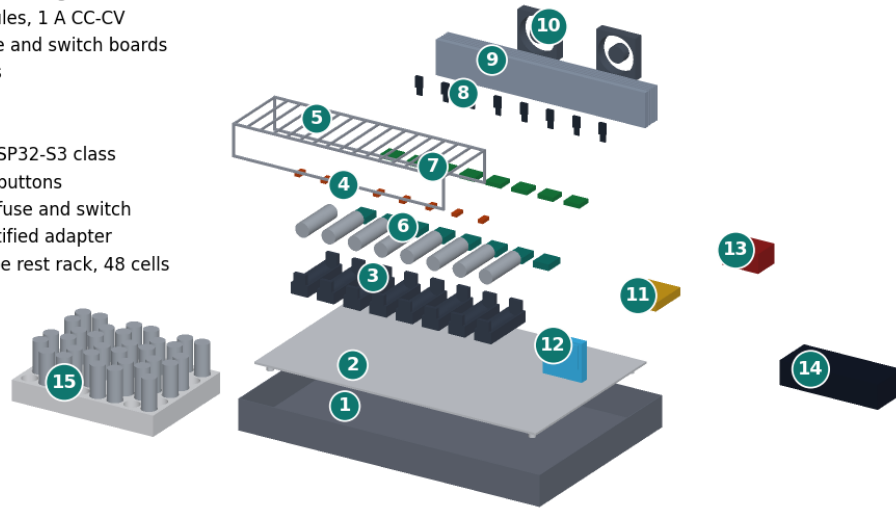


Figure 3. Exploded view. Numbers match Table 1 and `bom/bom.csv`. Grey cells are not supplied.

■ Grading rules

Table 2. Proposed grades (thresholds are estimates for review; all live in a plain configuration file). Proposed, awaiting Amish.

Grade	Capacity, share of rated	DC resistance (18650 default)	Other conditions	Suggested use
A	80 % or more	60 mΩ or less	Temperature rise 8 K or less at 1 A; self-discharge 50 mV or less in 14 days	Storage packs at 0.5C or less
B	65 % or more	100 mΩ or less	As A	Storage and lighting packs at 0.3C or less
C	50 % or more	150 mΩ or less	As A	Low-drain uses only: lights, small sensor nodes
Reject	Below 50 %	Above 150 mΩ	Or any failed condition, arrival voltage below 2.0 V, or visible damage	Battery recycling

Rated capacity comes from the cell model printed on the wrap, looked up in a cell table; if the model is unknown, the grade uses measured capacity only and the record says so. The 60 mΩ default for grade A is an assumption for a typical laptop 18650 in good health; it will be set per cell model at TRL 3.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3. Assumptions: a typical salvaged 18650 holds about 1.8 Ah as found (close to the 1.62 Ah mean reported by Olivero-Ortiz et al., 2026), arrives at about 30 % charge, has a mean voltage of about 3.65 V on discharge and about 3.9 V on charge; charger modules are about 85 % efficient and the adapter about 88 %; controller and fans draw about 3 W.

Table 3. First-order numbers.

Quantity	Estimate	Basis	Requirement
Time per cell	about 6.5 h (6 to 8 h)	Charge 1.6 h, rest 0.5 h, discharge 1.8 h, rest 0.25 h, recharge to 4.10 V 1.8 h, handling about 0.5 h; a 2.5 Ah cell takes about 8 h	
Throughput	about 12 cells per attended 10 h day; about 28 per 24 h if run overnight	8 channels, each started as soon as it is free	R5 met, thin margin
Cells for one 4S6P pack of grade A or B	about 36 candidates, about 3 attended days plus the 14-day rest	24 cells at about 66 % yield (Figure 2)	
Capacity accuracy	about ± 1 % after a one-time calibration, about ± 2 % without	25 m Ω shunt at 0.5 % tolerance, monitor gain error about 0.1 %	R2 met on estimate
Voltage accuracy	about ± 10 mV worst case before calibration, better after	Typical current-monitor data sheet figures	R3 at risk until checked
Resistance resolution	about 0.8 m Ω per count; about ± 3 m Ω repeatability expected	1.25 mV voltage step over a 1.5 A current step, 16 readings averaged, four-wire contacts	R4 met on estimate, unverified
Heat at full discharge load	about 29 W average, about 34 W peak	8 cells x 1.0 A x 3.65 V (4.2 V at start)	
Heatsink needed	1.1 K/W or better to keep MOSFET cases at 75 °C or less in a 35 °C room	34 W, about 2 K case-to-sink	R8 met on estimate with fans (about 0.5 to 0.8 K/W expected)
Adapter load, all channels charging	about 43 W of 60 W (about 71 %)	$8 \times 4.2 \text{ V} \times 1 \text{ A} / 0.85 + 3 \text{ W}$	
Energy per cell from the mains	about 18 Wh	(5.0 Wh first charge + 6.2 Wh recharge) / $0.85 + 2.4 \text{ Wh overhead, then} / 0.88$	Cost well under \$0.01 per cell at \$0.15/kWh
Cell heating at 1 A	about 0.1 W per cell, a few kelvin	1 A squared times about 0.1 Ω	An 8 K rise flags a fault
Size and mass	460 x 300 x 90 mm with guard; about 3.5 kg without adapter	Tray 2.0 kg, heatsink 0.6 kg, plate and modules 0.9 kg	R11 met
Parts cost	about \$159	Indicative prices, see <code>bom/bom.csv</code>	R12 met, about \$1 margin

■ Key design choices

- **Linear load, not energy recovery.** Discharge energy is burned in MOSFETs on one heatsink. Feeding it back to the 12 V bus or to a charging cell would save about 6.6 Wh per cell but adds a bidirectional converter per channel and costs more than the budget allows. Proposed, awaiting Amish; regenerative discharge is a possible later variant.

- **One cell per channel, eight channels.** Eight channels keep heat near 30 W and cost within budget while giving about 12 cells a day. A 16-channel version is an option for a busy repair shop. Recommendation: eight channels. Proposed, awaiting Amish.
- **Four-wire DC resistance, not 1 kHz AC impedance.** DC pulses need no signal generator and reflect how the cell behaves under load; AC readings from handheld meters are lower and not comparable. Recommendation: DC. Proposed, awaiting Amish.
- **Passive rest rack for self-discharge.** Holding cells in channels for 14 days would stall the grader, so cells rest in a rack and come back for a 10 s reading. Recommendation: 14 days (7 days is an option that catches only the worst cells). Proposed, awaiting Amish.
- **12 V DC input only.** A certified adapter keeps mains out of the unit, so builders never wire mains.
- **Local-first data.** Records stay on the microSD card and the local web page; no cloud account is needed. The CSV fields could map to ReflowEconomy's material passport at TRL 3. Proposed, awaiting Amish.
- **Lithium-ion first, LFP later.** The first release covers LCO, NMC and NCA cells at 4.20 V. LFP cells need a 3.65 V charger and a 2.5 V discharge limit; an LFP profile is an open question.

■ Relation to other lab projects

- **SwapCell.** The pitch names SwapCell as a user of rebuilt packs. SwapCell's current spec specifies new, matched 5 Ah 21700 cells carrying 10 A continuous each in a 13S2P pack. Typical laptop cells are rated for far less current, so graded laptop cells cannot meet the SwapCell reference pack as specified. A second-life SwapCell variant (for example a low-current storage pack for PowerBox, or a pack of graded power tool cells) needs a decision by Amish and agreement with the SwapCell project. Proposed, awaiting Amish.
- **CellGuard.** Packs rebuilt from graded LFP cells would use CellGuard; lithium-ion packs depend on CellGuard's proposed NMC profile, which is itself awaiting Amish.
- **ReflowEconomy.** CellCheck is the grading step for the battery stream in ReflowEconomy's micro-factory, where cells that cannot be reused are exported for industrial recycling.

■ Safety

Safety: Salvaged lithium-ion cells can have hidden damage and can overheat, vent flammable and toxic gas, and burn. Work on a non-flammable surface, with a smoke alarm and a Class D or large dry-powder extinguisher or a sand bucket within reach, and never leave the grader running unattended until Amish has decided whether and how overnight operation is allowed.

- **Pack disassembly.** Removing spot-welded nickel strip can short cells or tear wraps. Use insulated tools, cut one strip at a time, wear eye protection and gloves, and tape exposed terminals.
- **Damaged and deeply discharged cells.** Reject cells with any visible damage and cells below 2.0 V. Do not try to revive them.
- **Charging.** Each charger module limits voltage to 4.20 V by itself; the firmware adds temperature and time limits. A cell that passes 45 °C or rises 8 K above the bench is disconnected and flagged; the operator moves it to a metal container of sand once it is cool.
- **Single faults.** A load MOSFET that fails short would drain its cell far below 2.0 V; the cell must then be rejected. A hardware watchdog that removes all load gate drive when the controller stops is proposed (R9).

- **Heat.** The heatsink can reach about 70 °C; the guard and layout keep it away from the cells, and the fans run whenever a load is on.
- **Containment.** The steel tray, ceramic fibre liner and steel guard are meant to hold ejecta from one venting cell. This has not been checked and must not be relied on.
- **Transport and storage.** Store graded cells at about 3.7 V in a non-flammable container with terminals covered. Rejected cells go to a battery collection point, never to household waste.
- **Scope.** CellCheck is a research and prototype tool. Its grades do not certify any cell or pack as safe.

■ Open questions

- [] May graded cells ever go into SwapCell packs, and if so, which cell sources and current limits? Proposed, awaiting Amish.
- [] Can the grader run unattended overnight, and with what precautions (for example a fire-rated cabinet and a remote alarm)? Proposed, awaiting Amish.
- [] Grading thresholds, rest period and matching tolerance (Table 2 and item 7). Proposed, awaiting Amish.
- [] Should an LFP profile be in the first release? Proposed, awaiting Amish.
- [] Which partner should supply cells for first trials: a repair café, an e-bike repair shop or a collection point? Proposed, awaiting Amish.
- [] How does the grading record map onto ReflowEconomy's material passport? To be studied at TRL 3.